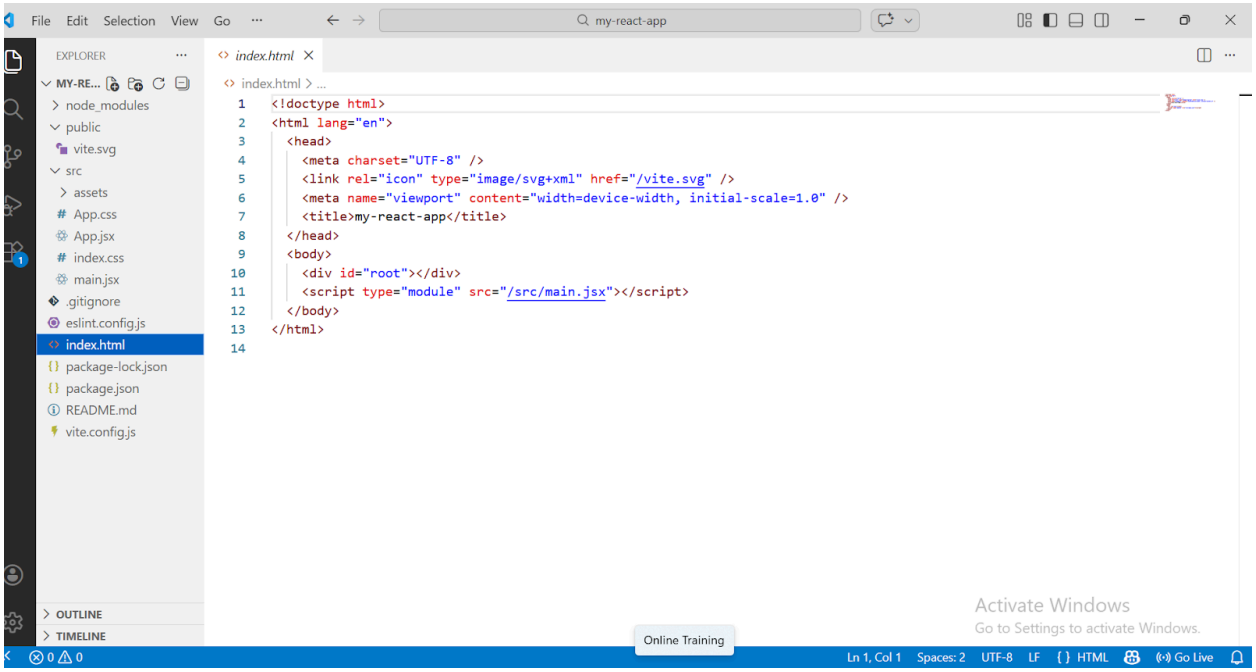
React Part 1:-
Components of react app:-
1.node modules
2.public
3.src

What is react?
React is a JavaScript library used to build user interfaces (UI) especially for web applications.
It was created by Facebook (now Meta).

SPA approach:-
Only ONE HTML page is loaded initially.
After that, React dynamically updates content without reloading the page.

Basic structure of react project:-



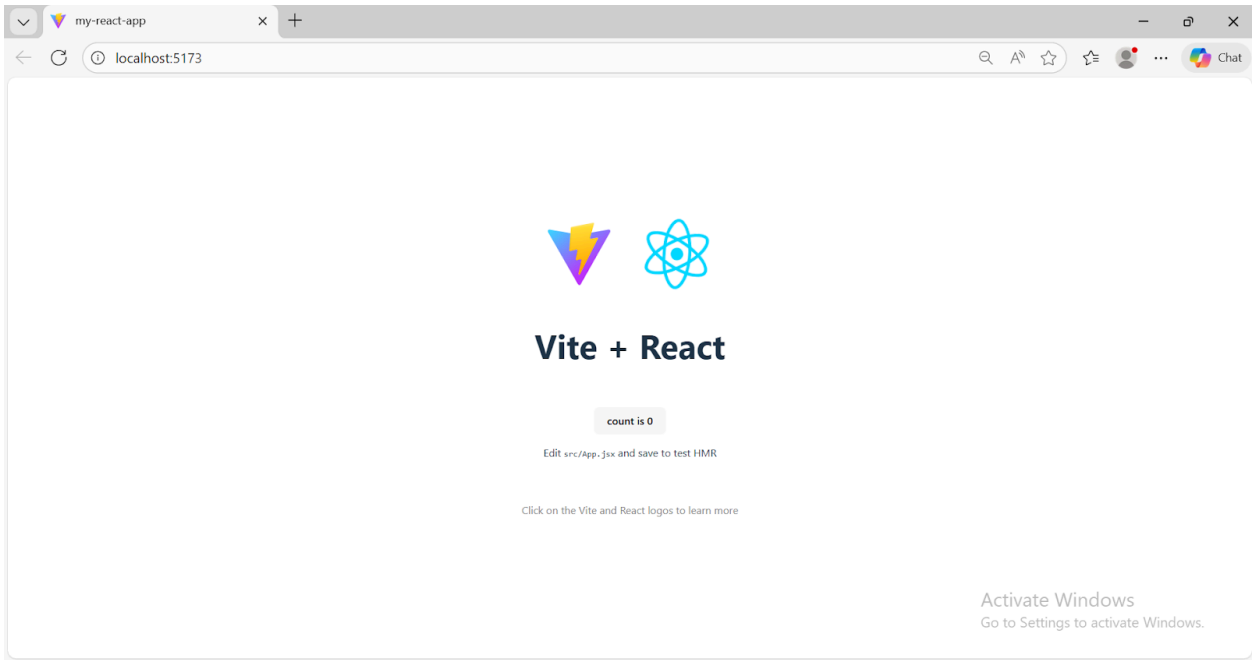
1.node_modules
This folder contains all the installed dependencies and packages required for the React application.It is automatically created when we run npm install and should not be modified manually.

2.public
This folder contains static files like index.html, images, and icons that are directly served to the browser.The index.html file inside this folder is the main HTML file where the React app gets rendered.

3. src
This folder contains the main source code of the React application such as components, CSS, and JavaScript files.It includes important files like App.js and index.js where the application logic and UI are written.

When we want to share our project to other we just share the src folder
The respective destination system installs the necessary packages and then runs the project in their system.

To run the project:-
1.open the terminal
2.Type the command npm start
3.Press Enter

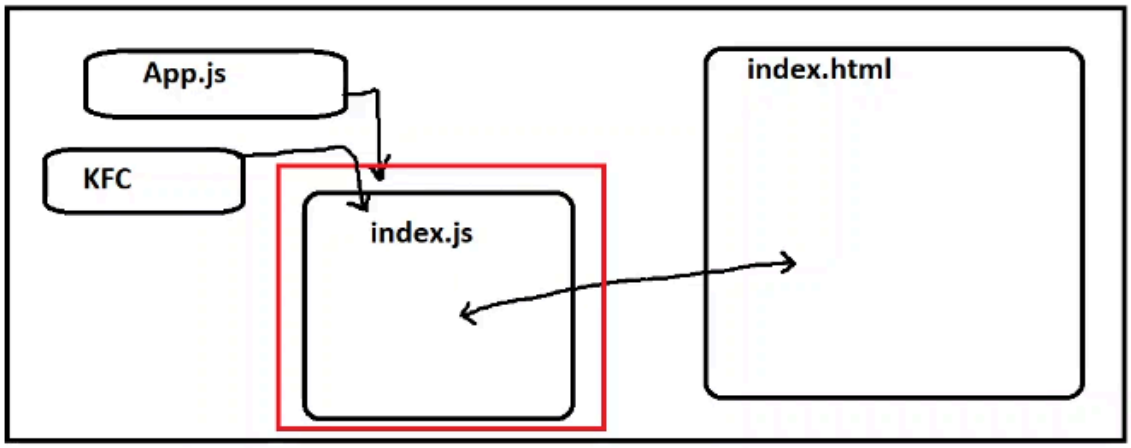


Always note that the component name starts with a capital letter.

React JSX:-
App.jsx

- App.jsx is the main component of a React application.
- It acts as the root component where all other components are created and organized.
- App.jsx is the central file that defines the main structure of the React application.
- It controls how different components are displayed and rendered on the screen.It combines multiple components and forms the complete user interface.
- It manages application-level logic such as state management and routing.

- App.jsx exports a component.This component is imported into main.jsx (Vite) or index.js (Create React App).
- React then renders the App component into the root DOM element.



getElementById() is a JavaScript method used to select an HTML element by its unique id. It allows you to access and modify that element's content, style, or behavior.

How javascript is diffrent from jsx?

JavaScript	JSX
Pure programming language	Syntax extension used in React
Used everywhere (Node, browser, etc.)	Used mainly in React
No HTML inside JS by default	Allows HTML-like syntax inside JS
Works directly in browser	Needs transpilation (Babel)

What is javascrit xml?
JavaScript is a programming language used to make websites interactive and dynamic in the browser.
XML (Extensible Markup Language) is a markup language used to store and transport structured data.

When people say “JavaScript XML,” they usually mean AJAX (Asynchronous JavaScript and XML), a technique that lets web pages update data without refreshing.

Why do we use <> (angular brackets) in app.jsx?

<>/</> are called react fragments.

```
return (
  <h1>Hello</h1>
  <p>World</p>
)
```

This is not allowed in react.
Instead we write:-

```
return (
  <div>
    <h1>Hello</h1>
    <p>World</p>
  </div>
)
```

Sometimes we dont want extra div in react.
So react gives us fragments

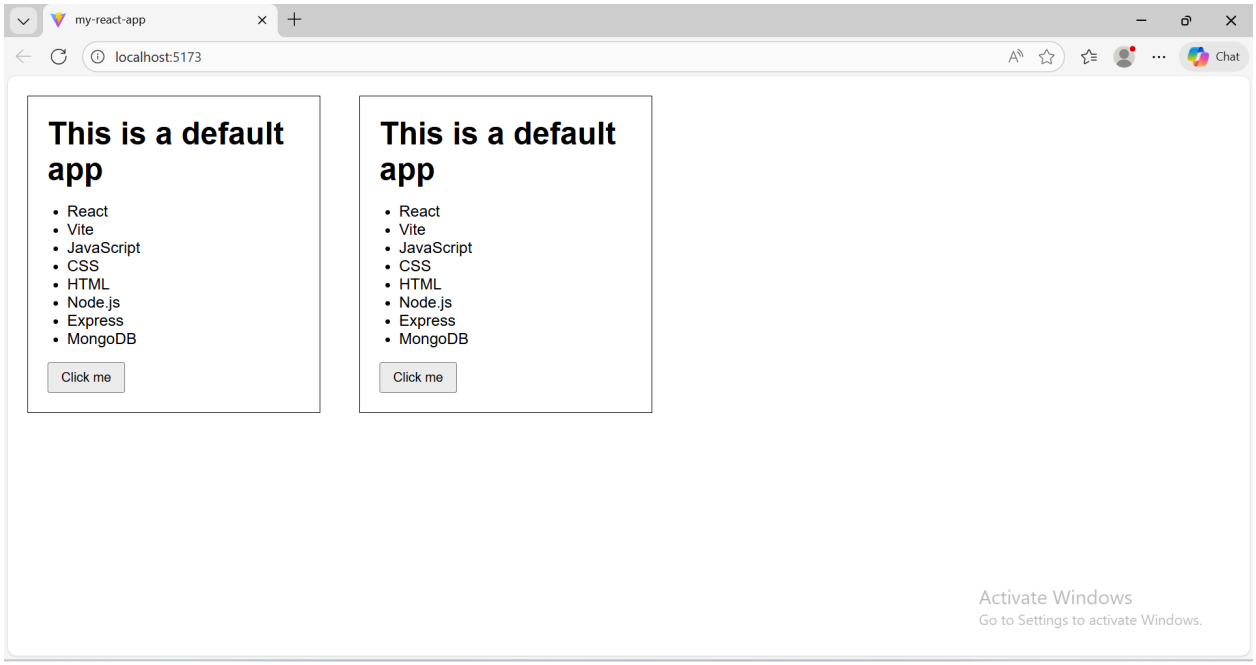
```
return (
  <>
    <h1>Hello</h1>
    <p>World</p>
  </>
)
```

If we wry=ite div <> /</>fragments are not needed and viceversa.

App.jsx

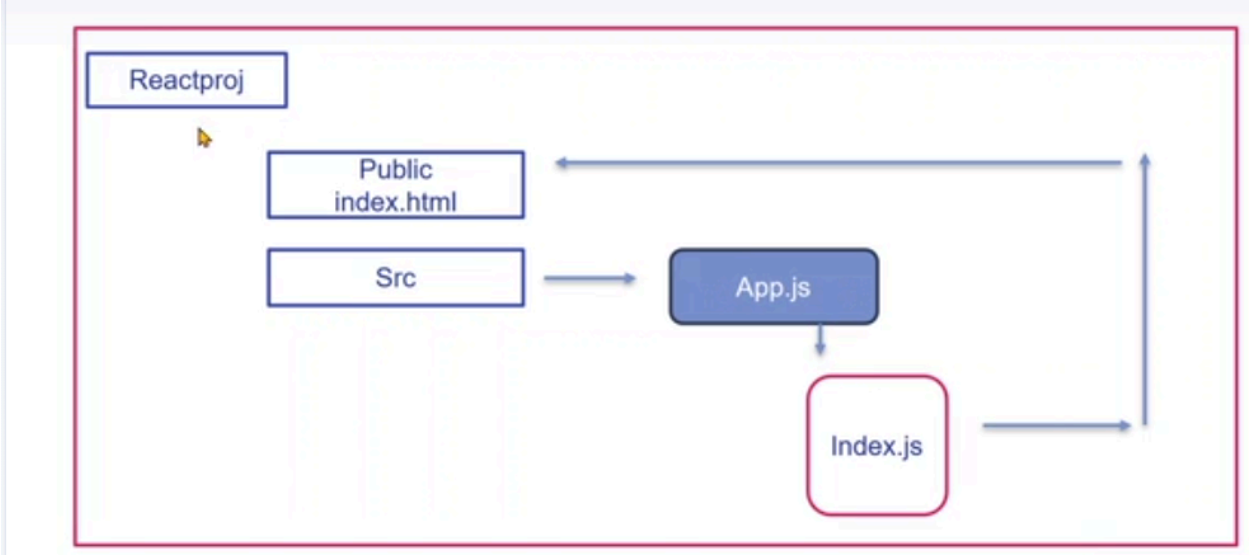
```
import { useState } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from '/vite.svg'
import './App.css'

function App() {
  return (
    <div className="container">
      <div className="block">
        <h1>This is a default app</h1>
```

```
        <ul>
          <li>React</li>
          <li>Vite</li>
          <li>JavaScript</li>
          <li>CSS</li>
          <li>HTML</li>
          <li>Node.js</li>
          <li>Express</li>
          <li>MongoDB</li>
        </ul>
        <button>Click me</button>
      </div>

      <div className="block">
        <h1>This is a default app</h1>
        <ul>
          <li>React</li>
          <li>Vite</li>
          <li>JavaScript</li>
          <li>CSS</li>
          <li>HTML</li>
          <li>Node.js</li>
          <li>Express</li>
          <li>MongoDB</li>
        </ul>
        <button>Click me</button>
      </div>
    </div>
  )
}


export default App
```



React Hooks:-
React Hooks are special functions that let you use React features (like state and lifecycle methods) inside functional components.

Basic structure of a React project :-



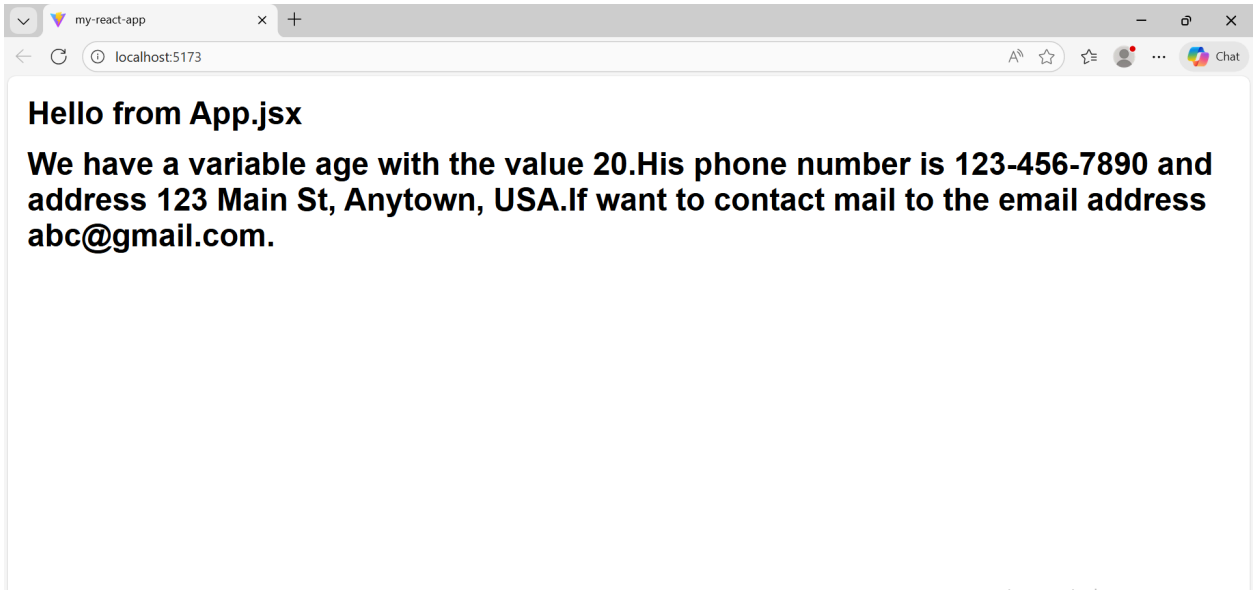
JSX=HTML+JS+Events+Data

What is useState in React?

useState is a React Hook that allows functional components to store and manage data (state).It lets your component remember values and update the UI whenever those values change.

App.jsx

```
import { useState } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from '/vite.svg'
import './App.css'

function App() {
  const [age,setAge] = useState(20)
  const [phone,setPhone]=useState('123-456-7890')
  const [email,setEmail]=useState('abc@gmail.com')
  const [address,setAddress]=useState('123 Main St, Anytown, USA')

  return (
    <>
    <h1>Hello from App.jsx</h1>
    <h1>We have a variable age with the value {age}.His phone number is
{phone} and address {address}.If want to contact mail to the email address
{email}.</h1>
</>
  )
}


export default App
```



React Array:-

In React, arrays are used to store multiple values and are commonly rendered using the map() method to display lists dynamically.

Hooks - class components

Array -function components

Scenario:-

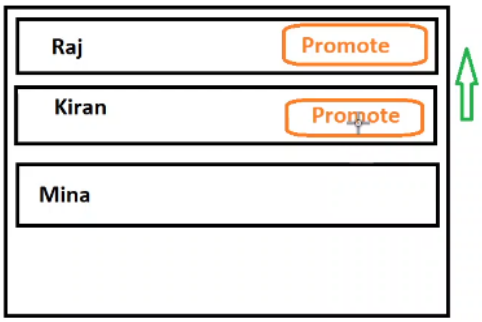
Create a react component to display the employees in the organization and implement logic to promote employees.

1.Create a component.

2.Create employee data.

3.Display the employee data in HTML data.

Before:-



Components name should begin with capital letter.

Promote.jsx

```
function Promote() {
    return(
     <h1>Employee Promotion Module</h1>
    )
}
export default Promote;
```

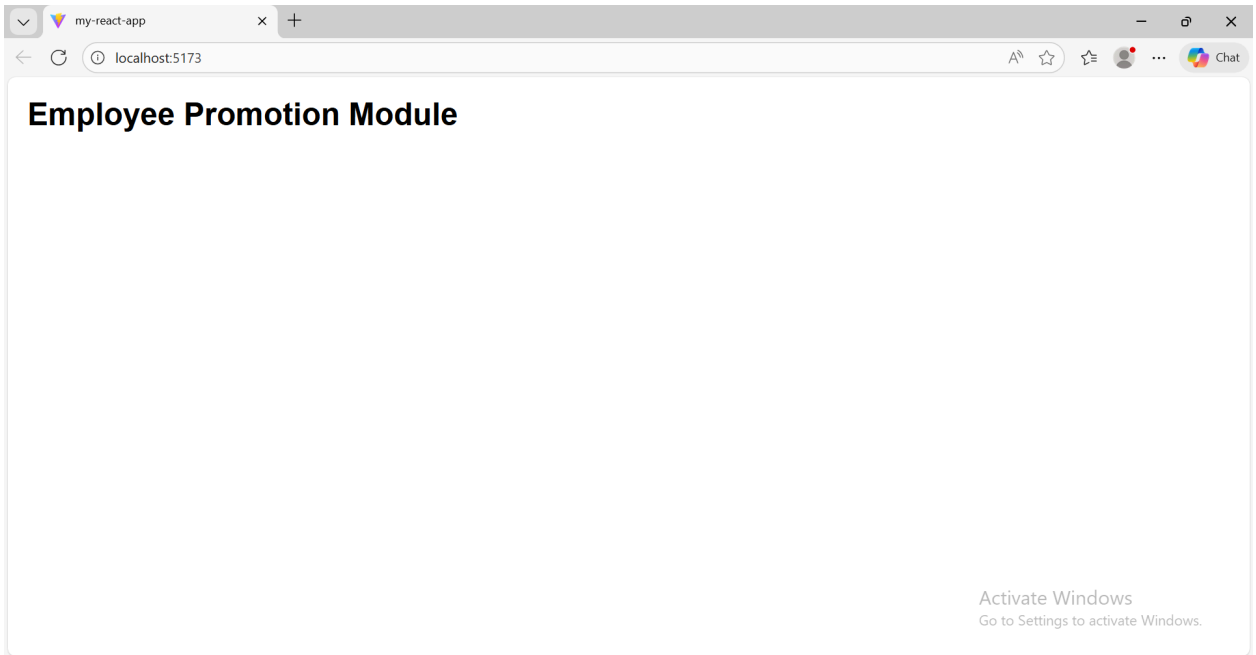
App.jsx

```
import { useState } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from '/vite.svg'
import './App.css'
import Promote from './components/Promote'

function App() {
  const [age,setAge] = useState(20)
  const [phone,setPhone]=useState('123-456-7890')
  const [email,setEmail]=useState('abc@gmail.com')
  const [address,setAddress]=useState('123 Main St, Anytown, USA')

  return (
 <>
       <div>
      <Promote/>
    </div>
</>
  )
}


export default App
```

Main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'
```

```
createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```



The data associated with a component is also called state.

React Array,Table
Map function:-

map() is used to loop through an array and create a new array.In React, we mainly use map() to display list data dynamically.

Example:-

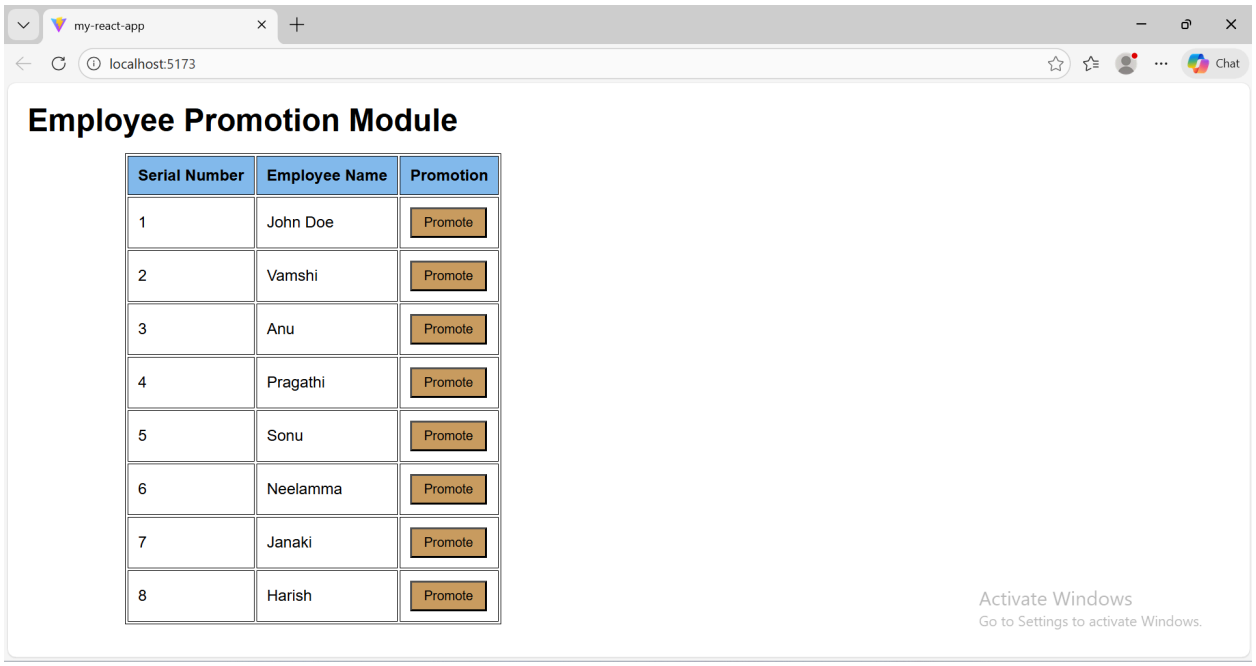
```
const numbers = [1, 2, 3];
const doubled = numbers.map(num => num * 2);
console.log(doubled);
// Output: [2, 4, 6]
```

Promote.jsx

```
import { useState } from "react";

function Promote() {
    const [employees, setEmployees] = useState([
        'John Doe',
        'Vamshi',
        'Anu',
        'Pragathi',
        'Sonu',
        'Neelamma',
        'Janaki',
        'Harish'
    ]);

    return (
        <>
            <h1>Employee Promotion Module</h1>
            <table>
                <thead>
                    <tr>
                        <th >Serial Number</th>
                        <th>Employee Name</th>
                        <th>Promotion</th>
                    </tr>
                </thead>
                <tbody>
                    {employees.map((emp, index) => (
                        <tr key={index}>
                            <td>{index + 1}</td>
                            <td>{emp}</td>
                            <td><button>Promote</button></td>
                        </tr>
                    ))}
                </tbody>
            </table>
        </>
    );
}
```

```
export default Promote;
```

App.css

```
/* /* Reset root container */
#root {
  margin: 0;
  padding: 0;
  text-align: left;
}

/* Simple spacing */
h1 {
  margin-bottom: 15px;
}

ul {
  margin-bottom: 15px;
  padding-left: 20px;
}

button {
  padding: 6px 12px;
  cursor: pointer;
}
.container {
  display: flex;
  gap: 40px;
}

.block {
  border: 1px solid black;
  padding: 20px;
  width: 300px;
}
table{
  margin-left: 100px;
  border:1px solid black;
}
th{
```

```
  background-color: rgb(131, 187, 236);
}
th,td{
  border:1px solid black;
  padding:10px;
}
button{
  background-color: rgb(201, 155, 98);
}
```